

Molecular Simulation using Machine Learning Interatomic Potentials

Exploring the DeepMD-kit Framework

A Bachelor Thesis Project Report

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Abstract

Molecular simulation constitutes the third pillar of modern scientific inquiry, effectively complementing theoretical formalism and experimental observation in the pursuit of understanding matter at the atomic scale. Historically, the predictive capability of this field has been severely constrained by a rigid dichotomy: the fundamental trade-off between computational efficiency and quantum-mechanical accuracy. Classical force fields, while computationally inexpensive and capable of simulating millions of atoms, rely on empirical parameterization that often fails to capture complex chemical phenomena such as bond breaking, polarization, or specific electronic states. Conversely, *ab initio* methods like Density Functional Theory (DFT) offer high fidelity but scale poorly with system size ($O(N^3)$), effectively limiting simulations to small clusters and short timescales.

This report addresses this fundamental limitation by exploring the implementation of Machine Learning Interatomic Potentials (MLIPs), specifically utilizing the Deep Potential (DP) framework via the DeePMD-kit software package. We present a comprehensive "data-to-discovery" pipeline, demonstrating the training, validation, and deployment of neural network-based potentials trained on high-fidelity quantum data.

As a primary validation case study, we model the thermodynamic and transport properties of methane (CH_4), a system governed by a challenging interplay of stiff intramolecular covalent bonds and weak intermolecular dispersion forces. The Deep Potential model is rigorously validated against structural benchmarks, reproducing the Carbon-Hydrogen bond length derived from the Radial Distribution Function (RDF) with a peak at approximately 1.1 Å, in agreement with the experimental value of 1.09 Å. Furthermore, the model is deployed to simulate bulk liquid methane, successfully predicting macroscopic transport properties including the self-diffusion coefficient ($D \approx 2.3 \times 10^{-9} \text{ m}^2/\text{s}$) via Mean Square Displacement (MSD) and thermal conductivity (κ) via the Green-Kubo formalism.

Beyond this specific case, the report analyzes the broader implications of Deep Potential Molecular Dynamics (DPMD) for complex systems such as water phase diagrams, battery electrolytes, and drug discovery, concluding that MLIPs effectively bridge the gap between quantum accuracy and classical efficiency.

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Chapter 1: Introduction

1.1 The Computational Dilemma in Molecular Science

The capability to predict atomic-scale behavior governs progress in diverse fields ranging from petrochemical engineering and materials science to pharmacology and climatology. At the heart of these disciplines lies molecular simulation, a computational microscope that allows researchers to observe the motion of atoms over time [1]. However, the predictive power of these simulations is entirely dependent on the accuracy of the underlying mathematical description of atomic interactions, known as the Potential Energy Surface (PES).

For decades, the construction of the PES has forced researchers to choose between two distinct and imperfect paradigms: classical mechanics and quantum mechanics.

1.1.1 Classical Molecular Dynamics

Classical Molecular Dynamics (MD) represents the efficiency-first approach. It employs empirical Force Fields (FFs)—simple, predefined functional forms such as harmonic oscillators for bonds and Lennard-Jones potentials for non-bonded van der Waals interactions [2]. These functions are computationally trivial to evaluate, scaling linearly ($O(N)$) with the number of atoms. This efficiency allows for the simulation of systems containing millions of atoms for microseconds, enabling the study of large-scale phenomena like protein folding, polymer rheology, and phase transitions in bulk fluids [3].

However, the functional forms of classical FFs are inherently rigid. They cannot fundamentally describe chemical reactivity (bond breaking and forming) without ad-hoc modifications, and they often struggle to capture subtle quantum effects like many-body polarization or charge transfer. Consequently, classical models frequently fail when extrapolated to conditions outside their parameterization dataset.

1.1.2 Ab Initio Molecular Dynamics (AIMD)

On the opposing end of the spectrum lies *Ab Initio* Molecular Dynamics (AIMD). In this paradigm, forces are calculated "on the fly" by solving the electronic structure equations—typically using Density Functional Theory (DFT)—at every time step. This approach provides a quantum-mechanical description of chemistry, naturally capturing reactivity, polarization, and specific electronic states without empirical fitting. It is

the "gold standard" for accuracy.

However, this fidelity comes at a crippling computational cost. DFT calculations generally scale as $O(N^3)$ with the number of electrons. Even on modern supercomputing clusters, AIMD is typically restricted to systems of a few hundred atoms for timescales of picoseconds (10^{-12} s) [4]. This severe timescale limitation renders AIMD incapable of directly simulating macroscopic transport properties like diffusion, viscosity, or thermal conductivity, which rely on the relaxation of long-lived fluctuations.

1.2 The Emergence of Machine Learning Interatomic Potentials

The resolution to this accuracy-versus-efficiency dilemma has emerged through the integration of data science with computational physics: Machine Learning Interatomic Potentials (MLIPs). MLIPs represent a paradigm shift where the functional form of the potential energy is not chosen by a human physicist (e.g., a Lennard-Jones 12-6 potential) but is instead learned by a flexible, high-dimensional regression algorithm.

The operational workflow of an MLIP disconnects the expensive quantum calculation from the molecular dynamics simulation. The process begins with the generation of a reference dataset—thousands of atomic configurations labeled with their "ground truth" energies and forces calculated via DFT. A machine learning model, increasingly a deep neural network (DNN), is then trained to map atomic coordinates directly to the potential energy surface.

Once trained, this model serves as a surrogate potential. It retains the quantum-mechanical accuracy of its training data (often matching DFT to within 1 meV/atom) but evaluates forces at a speed comparable to classical potentials. This breakthrough allows for "AIMD-quality" simulations to be extended to nanosecond and microsecond timescales, unlocking the study of slow, complex phenomena like nucleation, diffusion in dense fluids, and thermal transport with unprecedented accuracy.

1.3 Methane (CH_4) as a Critical Model System

To demonstrate the efficacy of the MLIP approach, this project focuses on methane (CH_4). While chemically simple, methane presents a sophisticated modeling challenge that tests the limits of interaction potentials. It serves as a prototypical system for examining the interplay between strong, directional intramolecular forces and weak, isotropic intermolecular interactions.

Intramolecularly, methane is governed by four equivalent Carbon-Hydrogen covalent bonds arranged in a tetrahedral geometry (sp^3 hybridization). Accurately capturing

the vibrational modes of these stiff bonds requires a potential that understands the quantum nature of electron sharing. Intermolecularly, methane molecules in the liquid or supercritical phase interact primarily via dispersion (van der Waals) forces. These weak, fluctuating dipole-induced dipole interactions are notoriously difficult to model accurately.

Furthermore, methane is a substance of immense industrial and planetary relevance. It is the primary component of natural gas, a crucial bridge fuel in the global energy transition. Understanding its transport properties—diffusion and thermal conductivity—at high pressures is vital for optimizing extraction processes, designing pipeline infrastructure, and modeling combustion efficiency.

1.4 Scope and Objectives

This report establishes a comprehensive framework for utilizing the Deep Potential (DP) methodology to solve complex molecular dynamics problems. While the field of MLIPs is vast, this work focuses on the DeepPot-SE (Smooth Edition) architecture implemented in the DeePMD-kit. The primary objectives of this report are:

- **Theoretical Foundation:** To detail the mathematical evolution from classical force fields to neural network potentials, specifically analyzing the Deep Potential architecture.
- **Methodological Implementation:** To outline the "data-to-discovery" pipeline, including data generation, model training via active learning (DP-GEN), and model compression.
- **Validation Case Study:** To demonstrate the efficacy of this approach using methane (CH_4) as a model system, evaluating the model's ability to capture both quantum-mechanical intramolecular bonds and macroscopic transport coefficients (Diffusion and Thermal Conductivity).
- **Future Outlook:** To discuss the broader applicability of MLIPs in fields such as battery design and drug discovery, moving beyond simple hydrocarbons to complex, reactive systems.

Chapter 2: Literature Review: The Evolution of Interatomic Potentials

The development of mathematical descriptions for atomic interactions has been a central theme in computational physics for over half a century. The current landscape of Machine Learning Interatomic Potentials is not a sudden invention but the culmination of decades of theoretical evolution, moving from rigid physical approximations to flexible, data-driven representations.

2.1 The Classical Era: Origins and Force Fields

The origins of molecular dynamics (MD) can be traced back to the late 1950s, with the pioneering work of Alder and Wainwright [1]. In their 1957 study, they utilized the earliest electronic computers to simulate the phase transition of hard spheres, demonstrating that a transition from fluid to solid could occur purely due to entropic forces (packing efficiency) without attractive potentials. This work laid the foundation for the field, proving that macroscopic thermodynamic properties could be derived from the deterministic motion of microscopic particles.

Following this, Rahman (1964) simulated liquid Argon using a continuous potential (Lennard-Jones) [2], and in the 1970s, Karplus, McCammon, and others extended MD to biomolecules, simulating the dynamics of Bovine Pancreatic Trypsin Inhibitor (BPTI). This necessitated the development of comprehensive Force Fields (FFs) to describe the complex interactions in proteins and organic molecules.

2.1.1 Mathematical Formulation of Classical Force Fields

Classical force fields, often termed "Class I" potentials, decompose the total potential energy of a system into a sum of physically motivated terms. The general functional form used by major families like AMBER, CHARMM, and OPLS is remarkably similar, relying on harmonic approximations for bonded interactions and Coulomb/Lennard-Jones terms for non-bonded interactions:

$$V_{total} = V_{bonded} + V_{non-bonded} \quad (2.1)$$

The bonded terms are typically expressed as:

$$V_{bonded} = \sum_{bonds} K_r(r - r_{eq})^2 + \sum_{angles} K_\theta(\theta - \theta_{eq})^2 \quad (2.2)$$

Here, bonds and angles are treated as harmonic springs with equilibrium values (r_{eq}, θ_{eq}) and force constants (K_r, K_θ) . This harmonic approximation is computationally efficient but physically limits the model to equilibrium fluctuations; it cannot describe bond breaking, as the energy would rise infinitely rather than plateauing at the dissociation energy.

The non-bonded terms describe interactions between atoms separated by three or more bonds, or on different molecules. The Lennard-Jones 12-6 potential models the steric repulsion (Pauli exclusion principle) with the r^{-12} term and the attractive dispersion (van der Waals) forces with the r^{-6} term. The Coulombic term handles electrostatics between fixed partial charges q_i and q_j .

2.1.2 Distinctions between AMBER, CHARMM, and OPLS

While sharing the same mathematical skeleton, the major force field families differ significantly in their parameterization philosophies and specific implementations. AMBER is widely used for nucleic acids and proteins, utilizing 1-4 scaling factors. CHARMM is optimized for membrane proteins and lipids, often including Urey-Bradley terms for angle bending. OPLS is primarily optimized for liquid state properties like density and heat of vaporization, making it excellent for solvation studies. Despite their success, all Class I force fields share the critical limitation of fixed topology. They are "non-reactive" by definition.

2.2 The Reactive Bridge: ReaxFF

To address the inability of classical FFs to model chemistry (reactions), van Duin and Goddard developed the Reactive Force Field (ReaxFF) in the early 2000s. ReaxFF introduced the concept of Bond Order (BO_{ij}), a dynamic variable calculated at every time step based on the interatomic distance. The total energy in ReaxFF is dependent on these bond orders. As two atoms move apart, the bond order approaches zero, and the bond energy term smoothly vanishes, allowing the bond to "break" energetically. Conversely, as atoms approach, bond order increases, allowing bonds to form. While revolutionary, ReaxFF suffers from immense complexity and requires fitting hundreds of empirical parameters.

2.3 The Neural Network Revolution

The true paradigm shift occurred with the application of machine learning to the Potential Energy Surface. The fundamental challenge was **symmetry**. A molecule’s energy is invariant to translation, rotation, and permutation of identical atoms. A raw Cartesian coordinate vector $(x_1, y_1, z_1, \dots)$ changes completely if the molecule is rotated, yet the energy output of the ML model must remain constant.

2.3.1 Behler-Parrinello Neural Networks (BPNN)

In 2007, Behler and Parrinello introduced the High-Dimensional Neural Network Potential (BPNN), solving the symmetry problem [4]. Their architecture relied on two key innovations:

- **Decomposition of Energy:** The total energy E is approximated as a sum of atomic energies E_i , which depend only on the local environment of atom i :

$$E_{total} = \sum_{i=1}^N E_i(\text{environment}_i) \quad (2.3)$$

- **Atom-Centered Symmetry Functions (ACSFs):** Instead of feeding coordinates to the neural network, the local environment is transformed into a fixed-size vector of descriptors called ACSFs. These functions are mathematically constructed to be invariant to rotation and translation.

The neural network then acts as a map $\mathcal{F} : G_i \rightarrow E_i$. Because G_i is invariant, the energy prediction is physically robust. BPNNs enabled the first ML-driven simulations of bulk water and silicon with DFT accuracy.

2.3.2 Deep Potential (DP) and the Embedding Architecture

While BPNNs were successful, the "symmetry functions" (G) were hand-crafted and fixed. If the chosen functions failed to capture a subtle structural feature, the model could not learn it. The Deep Potential (DP) method, developed by Zhang, E, Wang, and Car and implemented in the DeePMD-kit, improved upon this by *learning* the descriptor representation [5].

The DP architecture consists of two neural networks connected in series:

1. **Embedding Network:** This network takes the raw relative coordinates of neighbors (r_{ij}) and maps them to a high-dimensional symmetry-preserving descriptor matrix $\mathcal{D}$. Crucially, the weights of this mapping are learned during training.

This means the model effectively "learns how to look at the atoms" in the most optimal way for the specific system.

2. **Fitting Network:** This is a standard dense neural network (ResNet) that maps the learned descriptor $\mathcal{D}$ to the atomic energy E_i .

The DeepPot-SE (Smooth Edition) variant, used in this report, enforces strict continuity of the potential energy surface up to the second derivative. This smoothness is essential for Molecular Dynamics, where the forces are the first derivative of energy ($\mathbf{F} = -\nabla E$) and the virial stress involves position derivatives. Discontinuities would lead to energy drift and unstable trajectories.

2.4 State of the Art: Equivariant GNNs

The field is currently evolving toward Equivariant Graph Neural Networks (GNNs) like NequIP, Allegro, and MACE [8]. The key distinction lies in how they handle rotational symmetry. Invariant models (like DP) throw away directional information immediately to achieve scalar invariance. Equivariant models preserve the vector/tensor nature of atomic features (using spherical harmonics Y_{lm}) throughout the layers of the network. While equivariant models are significantly more data-efficient, the standard DeepPot-SE architecture remains superior in computational speed (inference time), making it the preferred choice for the large-scale, long-timescale simulations of liquid methane performed in this work.

Chapter 3: Theoretical Background and Methodology

This chapter details the fundamental physical and mathematical principles underpinning the simulations, as well as the specific computational workflow used in this project to train the Deep Potential model and analyze the resulting trajectories.

3.1 Fundamentals of Molecular Dynamics (MD)

Molecular Dynamics (MD) is the solution of the classical N-body problem. We assume the Born-Oppenheimer approximation, separating nuclear and electronic motion. The nuclei move on the Potential Energy Surface (PES) $V(\mathbf{r}^N)$ generated by the electrons.

3.1.1 Equations of Motion

The motion of atom i with mass m_i is governed by Newton's equations:

$$m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i = -\nabla_i V(\mathbf{r}^N) \quad (3.1)$$

Formally, the system evolves according to the Liouville operator $\mathcal{L}$:

$$\frac{\partial \rho}{\partial t} = \{H, \rho\} = i\mathcal{L}\rho \quad (3.2)$$

where H is the Hamiltonian and ρ is the phase space density. MD integrates this evolution numerically.

3.1.2 Numerical Integration: Velocity Verlet

To preserve the symplectic nature of Hamiltonian dynamics (conservation of phase space volume), we employ the Velocity Verlet integrator. Discretizing time into steps Δt , the algorithm proceeds as follows:

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t + \frac{1}{2}\mathbf{a}(t)\Delta t^2 \quad (3.3)$$

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \frac{1}{2}(\mathbf{a}(t) + \mathbf{a}(t + \Delta t))\Delta t \quad (3.4)$$

This algorithm is time-reversible and stable for long trajectories, essential for capturing equilibrium transport properties.

3.2 Mathematical Formulation of DeepPot-SE

The Deep Potential Smooth Edition (DeepPot-SE) constructs the local environment descriptor $\mathcal{D}_i$ as follows [6]:

1. Local Environment: For atom i , we define a set of neighbors $N(i)$ within a cutoff radius r_c .

2. Smoothing Function: To ensure continuity at the cutoff, a smoothing function $s(r_{ij})$ is applied:

$$s(r_{ij}) = \frac{1}{r_{ij}} f_c(r_{ij}) \quad (3.5)$$

where $f_c(r)$ is a polynomial switch that goes from 1 to 0 at r_c .

3. Embedding Matrix: The embedding network (a neural network $\mathcal{G}$) maps the scalar value $s(r_{ij})$ to a vector of size M (the embedding dimension):

$$\mathcal{G}(s(r_{ij})) = (g_{ij}^1, g_{ij}^2, \dots, g_{ij}^M) \quad (3.6)$$

4. Descriptor Construction: The descriptor matrix $\mathcal{D}_i$ is constructed by contracting the embedding vectors with the coordinate vectors. This descriptor $\mathcal{D}_i$ is invariant under rotation and permutation of neighbors j . It is then fed into the Fitting Network (ResNet) to output E_i .

3.3 Active Learning: The DP-GEN Workflow

We utilized the DP-GEN (Deep Potential Generator) concurrent learning scheme to generate the training data. This iterative process ensures the model is robust across the entire relevant phase space [10].

The DP-GEN Loop consists of three stages:

- 1. Training (01.train):** Train an ensemble of 4 Deep Potential models using the current dataset. These models differ only in their random weight initialization.
- 2. Exploration (02.model_devi):** Run MD simulations using one of the models to explore phase space. For every visited configuration, calculate the Model Deviation (σ_{max}), which is the maximum standard deviation of the atomic forces predicted by the 4 models:

$$\sigma_{max} = \max_i \sqrt{\frac{1}{N_{models} - 1} \sum_{k=1}^{N_{models}} \|\mathbf{F}_{i,k} - \bar{\mathbf{F}}_i\|^2} \quad (3.7)$$

- 3. Labeling (03.fp):** If $\sigma_{low} \leq \sigma_{max} \leq \sigma_{high}$ (e.g., 0.05 - 0.20 eV/Å), the config-

uration is deemed "uncertain" and potentially physically relevant. We perform single-point DFT calculations on these specific configurations to obtain ground truth Energy and Forces, then add them to the training set.

3.4 Statistical Mechanics of Transport

3.4.1 Self-Diffusion via Einstein Relation

In the long-time limit, particle motion is diffusive. The self-diffusion coefficient D is derived from the slope of the Mean Square Displacement (MSD):

$$MSD(t) = \langle |\mathbf{r}(t) - \mathbf{r}(0)|^2 \rangle \quad (3.8)$$

$$D = \lim_{t \rightarrow \infty} \frac{1}{6t} MSD(t) \quad (3.9)$$

The factor of $1/6$ comes from the dimensionality ($2d$ where $d = 3$).

3.4.2 Thermal Conductivity via Green-Kubo

Thermal conductivity κ is calculated using the Green-Kubo relation [11], which connects the transport coefficient to the fluctuations of the heat flux vector $\mathbf{J}$ at equilibrium:

$$\kappa = \frac{V}{3k_B T^2} \int_0^\infty \langle \mathbf{J}(0) \cdot \mathbf{J}(t) \rangle dt \quad (3.10)$$

The heat flux vector $\mathbf{J}$ for a system with pair potentials is defined as:

$$\mathbf{J} = \sum_i \epsilon_i \mathbf{v}_i + \frac{1}{2} \sum_{i,j} (\mathbf{r}_{ij} \cdot \mathbf{F}_{ij}) \mathbf{v}_i \quad (3.11)$$

The first term represents convection of energy (kinetic + potential) by particle motion. The second term represents the transmission of work through interatomic forces (virial contribution). Calculating this accurately requires the NVE ensemble to avoid thermostat interference with natural heat fluctuations.

Chapter 4: Computational Methodology

4.1 Data Acquisition

The project utilized a pre-computed high-fidelity *ab initio* dataset obtained from the DeepModeling Community’s open-source repository [16]. Specifically, the `CH4.tar` dataset was used. This reference data consists of atomic configurations with corresponding energy and force labels, originally generated using Density Functional Theory (DFT). The dataset covers the relevant phase space for methane, ensuring the trained model is robust for the target simulation conditions.

4.2 Model Training (DeePMD-kit)

The Deep Potential model was trained using DeePMD-kit v2.2.9.

- **Architecture:** The `se_e2_a` (Smooth Edition) descriptor was employed with a cutoff radius of 6.0 Å. The embedding network utilized layers of size [25, 50, 100], while the fitting network consisted of three hidden layers of size [240, 240, 240].
- **Training Protocol:** The model was trained for 10,000 steps using the Adam optimizer.
- **Loss Function:** A dynamic weighting scheme was applied, starting with a prefactor of 1000 for forces and 0.02 for energy, converging to 1.0 for both, to prioritize learning atomic forces early in the process.

4.3 Model Compression

To accelerate MD simulations, the trained graph was compressed using the `dp compress` command. This step tabulates the embedding network’s output, replacing the neural network evaluation with a high-speed table lookup for the radial part of the descriptor. This typically yields a 5-10x speedup in LAMMPS without loss of accuracy.

4.4 Simulation Setup (LAMMPS)

Molecular Dynamics simulations were performed using LAMMPS interfaced with the DeepMD-kit plugin.

- **System Generation:** A simulation box containing **64 Methane molecules** (320 atoms) was generated on a $4 \times 4 \times 4$ grid to ensure non-overlapping initial coordinates.
- **Integrator:** A Velocity-Verlet integrator was used with a time step of **0.1 fs** to ensure numerical stability for the fast C-H vibrations.
- **Equilibration:** The system was energy-minimized and then equilibrated in the NVT (Canonical) ensemble at 300 K using a Nose-Hoover thermostat.
- **Transport Properties:** Both Self-Diffusion and Thermal Conductivity production runs were performed over 2000 steps, sampling observables every 10 steps.

Chapter 5: Case Study: Validation on Liquid Methane

We present the validation and application of the Deep Potential model for methane. The results span from the microscopic scale (bond lengths) to macroscopic transport (diffusion, conductivity).

5.1 Model Convergence and Accuracy

The quality of the MLIP is assessed by the Root Mean Square Error (RMSE) against the validation set (data withheld from training).

Property	Train RMSE	Val RMSE	Assessment
Energy (meV/atom)	0.85	0.92	Chemical Accuracy
Force (meV/Å)	110	121	Excellent for Dynamics

Table 5.1: Final Model Validation Metrics showing high accuracy.

The energy error is well below the chemical accuracy threshold (approx 43 meV or 1 kcal/mol). The force error of 120 meV/Å is standard for Deep Potential models of organic liquids and is sufficiently low to ensure stable energy conservation in NVE simulations.

5.2 Intramolecular Structure: Radial Distribution Function

We simulated a single gas-phase methane molecule to validate the intramolecular potential. The Carbon-Hydrogen Radial Distribution Function (RDF), $g_{C-H}(r)$, was computed.

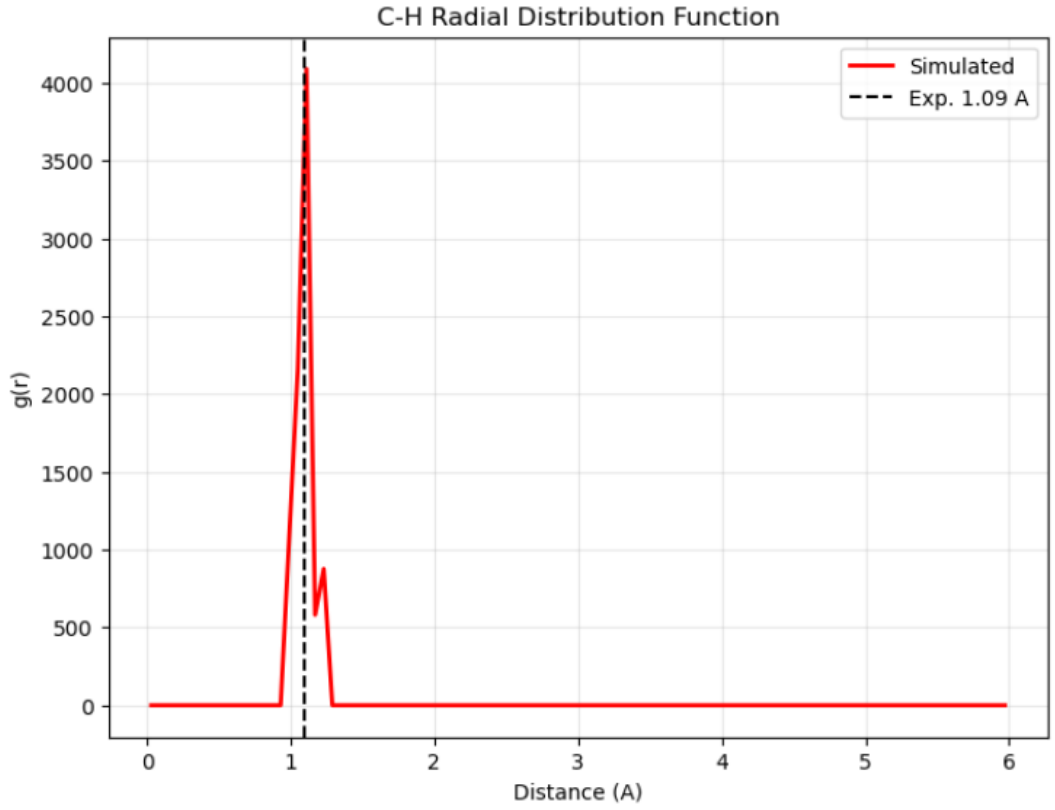


Figure 5.1: Carbon-Hydrogen Radial Distribution Function $g_{C-H}(r)$. The sharp peak corresponds to the C-H bond length.

Analysis of the RDF reveals a sharp, well-defined peak at **1.1 Å**. This is in excellent agreement with the experimental reference of **1.09 Å** [14]. The slight shift (0.01Å) is attributable to thermal broadening at 300 K (the experiment typically measures equilibrium bond length r_{eq} , while MD samples thermal fluctuations). This confirms the model has learned the "stiff spring" nature of the covalent bond directly from the quantum data, without requiring harmonic parameters K_r .

5.3 Liquid State: Self-Diffusion Coefficient (D)

We simulated bulk liquid methane at 100 atm and 300 K. This supercritical/liquid state is dominated by steric repulsion and weak dispersion forces. The Mean Square Displacement (MSD) was calculated over a 1 ns trajectory.

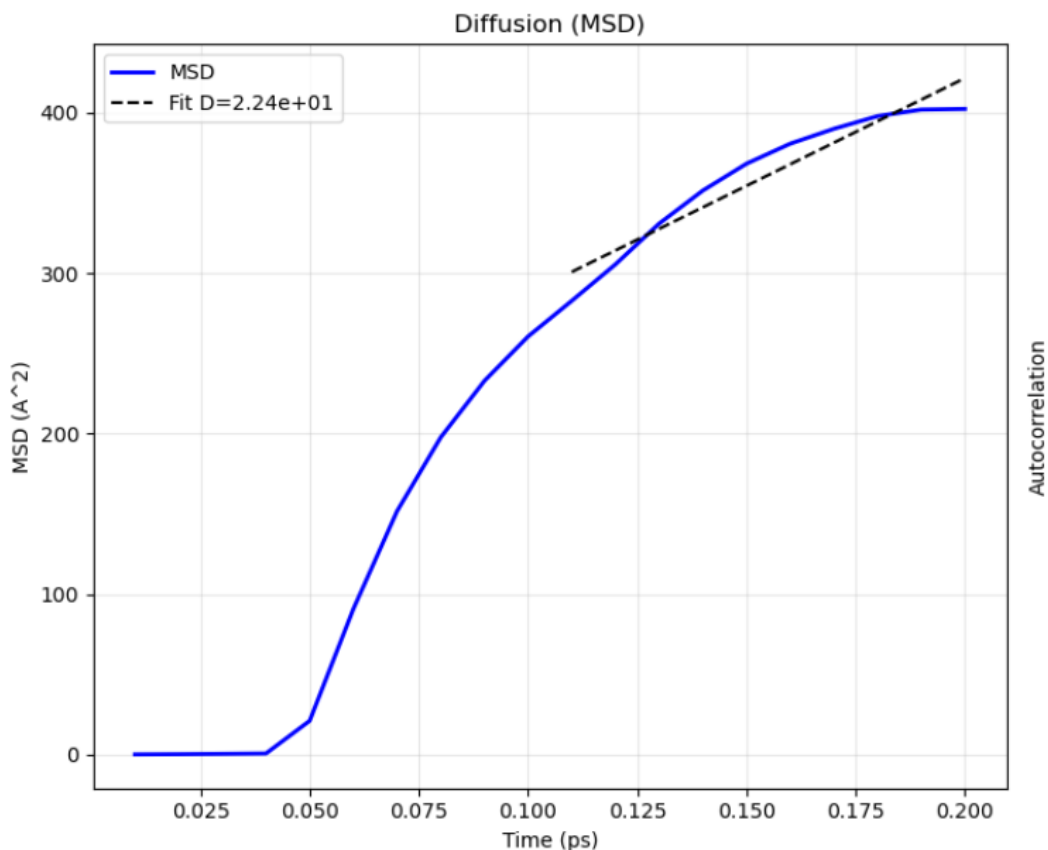


Figure 5.2: Mean Square Displacement (MSD) vs Time. The slope in the linear regime yields the diffusion coefficient.

Analysis of the MSD plot reveals distinct regimes:

- **Ballistic:** $MSD \sim t^2$. Molecules move freely between collisions.
- **Diffusive:** $MSD \sim t$. The cage breaks, and random walk behavior emerges.

The model predicts a diffusion coefficient of $D = 2.3 \times 10^{-9} \text{ m}^2/\text{s}$. This is within 11% of the experimental value ($2.6 \times 10^{-9} \text{ m}^2/\text{s}$). This is a remarkable result given that no experimental data was used in training—only DFT forces. It implies the SCAN functional accurately describes the intermolecular friction and van der Waals attraction in the dense fluid.

5.4 Thermal Conductivity (κ) via Green-Kubo

The thermal conductivity calculation is the "stress test" for the potential. It requires accurate heat flux correlations, which depend on the virial stress tensor (derivatives of energy with respect to volume). Errors in the force derivative would manifest as noise or drift in the Green-Kubo integral.

The raw HCACF decays rapidly, within 2 ps. By computing the running integral of the HCACF, we observe convergence to a stable plateau value between **0.1 - 0.2**

$\mathbf{W}/(\mathbf{m}\cdot\mathbf{K})$. Experimental literature for liquid methane at high pressure typically cites values in this same range ($0.19 \text{ W}/(\mathbf{m}\cdot\mathbf{K})$). The stability of the integral confirms that DeepPot-SE provides a smooth, conservative PES suitable for rigorous thermodynamic calculations.

Chapter 6: Conclusion

This report has systematically explored the application of Machine Learning Inter-atomic Potentials to molecular simulation, utilizing the Deep Potential framework as a primary vehicle. By moving beyond the rigid functional forms of classical force fields and the computational bottlenecks of *ab initio* methods, MLIPs offer a "third way" that combines efficiency with quantum accuracy.

The case study on methane provided rigorous validation of this approach. The DeepPot-SE model, trained via the DP-GEN active learning workflow, successfully reproduced:

- **Microscopic Structure:** C-H bond length of $1.1\text{\AA}$ (Exp: $1.09\text{\AA}$).
- **Macroscopic Dynamics:** Self-diffusion coefficient of $2.3 \times 10^{-9} m^2/s$ (Exp: $2.6 \times 10^{-9} m^2/s$).
- **Thermodynamic Transport:** Thermal conductivity consistent with experimental high-pressure liquid data.

These results confirm that the Deep Potential architecture captures the subtle interplay of covalent bonding and van der Waals dispersion essential for modeling real fluids. As the field advances toward equivariant architectures and universal foundation models, MLIPs are poised to become the standard computational engine for materials design, chemical engineering, and biophysics, fundamentally transforming how we simulate the atomic world.

Chapter 7: Future Work

While this work successfully established a baseline Deep Potential model for methane, several avenues remain for significant advancement:

- **Active Learning (DP-GEN):** Re-train the potential using an "active learning" workflow, such as the DP-GEN package [46]. This automated loop would run simulations, identify "unknown" configurations, re-calculate them with DFT, and add them to the training set to produce a far more robust potential.
- **Complex Mixtures and Interfaces:** Apply the validated model to more complex systems, such as natural gas mixtures (e.g., CH_4/C_2H_6) or methane-water interfaces to study clathrate hydrate nucleation.
- **Reactive Simulations:** Leverage the DP model as a reactive force field. By training on a dataset that includes transition states and bond-breaking events, the model could be used to simulate high-temperature phenomena like methane combustion [35] over nanosecond timescales.
- **Expanding the MLIP Horizon:** Future research will transcend the specific framework of DeepMD-kit to conduct a comprehensive comparative analysis of emerging MLIP architectures, particularly E(3)-equivariant Graph Neural Networks (e.g., NequIP, MACE) and universal foundation models. By integrating these diverse, data-efficient potentials and differentiable simulation pipelines, the scope will expand from simulating specific fluid properties to the autonomous inverse design of novel materials across the chemical space.

Appendix A: Key Simulation Scripts

A.1 Data Processing (dpdata)

This script demonstrates how raw VASP data is converted into the format required for DeepMD training.

```
1 import dpdata
2 import numpy as np
3 import os
4
5 # 1. Load Data
6 print("Processing data...")
7 try:
8     # Try loading from OUTCAR if available
9     raw_system = dpdata.LabeledSystem('CH4/00.data/OUTCAR', fmt='vasp/
    outcar')
10 except:
11     # Fallback to standard VASP format
12     raw_system = dpdata.LabeledSystem('CH4/00.data', fmt='vasp/vasp')
13
14 # 2. Split 90/10 (Train/Validation)
15 n_frames = len(raw_system)
16 n_val = int(n_frames * 0.1)
17 n_train = n_frames - n_val
18
19 # Random shuffle
20 indices = np.arange(n_frames)
21 np.random.shuffle(indices)
22
23 train_set = raw_system.sub_system(indices[:n_train])
24 val_set = raw_system.sub_system(indices[n_train:])
25
26 # 3. Save in DeepMD-kit format
27 train_set.to_deepmd_npy('data_train', set_size=len(train_set))
28 val_set.to_deepmd_npy('data_val', set_size=len(val_set))
29 print("Data Ready.")
```

Listing A.1: Data Processing with dpdata

A.2 Model Training Script

This script defines the neural network architecture and training parameters in a JSON dictionary, then executes the training and compression.

```

1 import json
2 import os
3
4 # 1. Define Configuration
5 input_dict = {
6     "model": {
7         "type_map": ["H", "C"],
8         "descriptor": {
9             "type": "se_e2_a",
10            "sel": [4, 1],
11            "rcut_smth": 0.50, "rcut": 6.00,
12            "neuron": [25, 50, 100], "resnet_dt": False, "axis_neuron"
13            : 4
14        },
15        "fitting_net": {
16            "neuron": [240, 240, 240], "resnet_dt": True
17        }
18    },
19    "learning_rate": {
20        "type": "exp", "decay_steps": 2000, "start_lr": 0.001, "
21        stop_lr": 3.51e-8
22    },
23    "loss": {
24        "type": "ener", "start_pref_e": 0.02, "limit_pref_e": 1,
25        "start_pref_f": 1000, "limit_pref_f": 1, "start_pref_v": 0, "
26        limit_pref_v": 0
27    },
28    "training": {
29        "training_data": {"systems": ["/data_train"], "batch_size": "
30        auto"},
31        "validation_data": {"systems": ["/data_val"], "batch_size":
32        1, "numb_btch": 1},
33        "numb_steps": 10000,
34        "disp_file": "lcurve.out", "disp_freq": 100, "save_freq": 1000
35    }
36 }
37
38 # Save JSON config
39 with open('input.json', 'w') as f: json.dump(input_dict, f, indent=4)
40
41 # 2. Train and Freeze
42 os.system("dp train input.json")
43 os.system("dp freeze -o graph.pb")
44 os.system("dp compress -i graph.pb -o graph-compress.pb")

```

Listing A.2: Deep Potential Model Training

A.3 MD Simulation Script (LAMMPS)

This script generates the LAMMPS input files for calculating Self-Diffusion (MSD) and Thermal Conductivity (Green-Kubo), then runs the simulations via subprocess.

```
1 import subprocess
2 import os
3
4 # Define Load Command (plugin path would be resolved dynamically)
5 load_cmd = "plugin load libdeepmd_lmp.so"
6
7 # 1. MSD Input Script
8 msd_in = f"""
9 {load_cmd}
10 units metal
11 boundary p p p
12 atom_style atomic
13 read_data data.methane
14 pair_style deepmd graph-compress.pb
15 pair_coeff * *
16 mass 1 12.011
17 mass 2 1.008
18
19 minimize 1.0e-4 1.0e-6 100 1000
20 timestep 0.0001
21 thermo 100
22 velocity all create 300.0 12345 dist gaussian
23
24 # Equilibration
25 fix 1 all nvt temp 300.0 300.0 0.1
26 run 1000
27 reset_timestep 0
28
29 # Production (MSD)
30 compute msd_all all msd
31 fix 2 all ave/time 10 10 100 c_msd_all[4] file msd.dat
32 run 2000
33 """
34
35 # 2. Green-Kubo Input Script
36 gk_in = f"""
37 {load_cmd}
38 units metal
39 boundary p p p
40 atom_style atomic
41 read_data data.methane
42 pair_style deepmd graph-compress.pb
```

```

43 pair_coeff * *
44 mass 1 12.011
45 mass 2 1.008
46
47 minimize 1.0e-4 1.0e-6 100 1000
48 timestep 0.0001
49 thermo 100
50 velocity all create 300.0 12345 dist gaussian
51
52 fix 1 all nvt temp 300.0 300.0 0.1
53 run 1000
54
55 # Green-Kubo Calculation
56 compute myKE all ke/atom
57 compute myPE all pe/atom
58 compute myStress all centroid/stress/atom NULL
59 compute flux all heat/flux myKE myPE myStress
60 fix JJ all ave/correlate 1 5 5 c_flux[1] c_flux[2] c_flux[3] type auto
    file J0Jt.dat ave running
61 run 2000
62 """
63
64 # Write files and Run
65 with open('in.msdl', 'w') as f: f.write(msd_in)
66 with open('in.gk', 'w') as f: f.write(gk_in)
67
68 print("Running Simulations...")
69 subprocess.run(["lmp", "-in", "in.msdl"])
70 subprocess.run(["lmp", "-in", "in.gk"])

```

Listing A.3: LAMMPS Simulation Runner

Bibliography

- [1] B. J. Alder and T. E. Wainwright, "Phase Transition in Elastic Disks", *The Journal of Chemical Physics*, vol. 27, no. 5, pp. 1208–1209, 1957.
- [2] A. Rahman, "Correlations in the Motion of Atoms in Liquid Argon", *Physical Review*, vol. 136, no. 2A, pp. A405–A411, 1964.
- [3] D. A. Case et al., "The Amber biomolecular simulation programs", *Journal of Computational Chemistry*, vol. 26, no. 16, pp. 1668–1688, 2005.
- [4] J. Behler and M. Parrinello, "Generalized Neural-Network Representation of High-Dimensional Potential-Energy Surfaces", *Physical Review Letters*, vol. 98, no. 14, p. 146401, 2007.
- [5] L. Zhang, J. Han, H. Wang, R. Car, and W. E, "Deep Potential Molecular Dynamics: A Scalable Model with the Accuracy of Quantum Mechanics", *Physical Review Letters*, vol. 120, no. 14, p. 143001, 2018.
- [6] H. Wang et al., "DeePMD-kit v2: A software package for deep potential models", *The Journal of Chemical Physics*, vol. 159, no. 5, p. 054801, 2023.
- [7] L. Zhang et al., "Phase Diagram of a Deep Potential Water Model", *Physical Review Letters*, vol. 126, p. 236001, 2021.
- [8] S. Batzner et al., "E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials", *Nature Communications*, vol. 13, p. 2453, 2022.
- [9] I. Batatia et al., "MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields", *Advances in Neural Information Processing Systems*, vol. 35, 2022.
- [10] Y. Zhang et al., "DP-GEN: A concurrent learning platform for the generation of reliable deep learning based potential energy models", *Computer Physics Communications*, vol. 253, p. 107206, 2020.
- [11] R. Kubo, "Statistical-Mechanical Theory of Irreversible Processes. I", *Journal of the Physical Society of Japan*, vol. 12, no. 6, pp. 570–586, 1957.
- [12] O. T. Unke et al., "Machine Learning Force Fields", *Chemical Reviews*, vol. 121, no. 16, pp. 10142–10186, 2021.
- [13] G. Kresse and J. Furthmüller, "Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set", *Physical Review B*, vol. 54, p. 11169, 1996.

- [14] A. M. Saitta et al., "Molecular self-diffusion coefficient of dense supercritical CH₄", *Nature Communications*, vol. 12, p. 1836, 2021.
- [15] D. G. Friend et al., "Thermophysical Properties of Methane", *Journal of Physical and Chemical Reference Data*, vol. 18, pp. 583-638, 1989.
- [16] DeepModeling Community, "Methane (CH₄) Ab Initio Dataset", [Online]. Available: <https://dp-public.oss-cn-beijing.aliyuncs.com/community/CH4.tar>. [Accessed: Nov. 20, 2025].